



ADITYA UNIVERSITY

MCA – I Semester End Examinations Regular – Dec 2024

OBJECT ORIENTED SOFTWARE ENGINEERING**Time: 3 hours****Max. Marks: 50****Answer ONE question from each unit****All Questions Carry Equal Marks****All parts of the questions must be answered at one place only****UNIT-I**

- 1 a Describe the RAD model and discuss its advantages and disadvantages? L2 CO1 [5M]
b What the significance of inheritance in object oriented software engineering and explain how does it facilitate polymorphism? L2 CO1 [5M]

(OR)

- 2 a What is meant by software development life cycle process and explain various types of models? L2 CO1 [5M]
b Define class and objects in object-oriented programming and explain with examples? L2 CO1 [5M]

UNIT-II

- 3 a Explain about system's functional and non-functional requirements using examples? L2 CO2 [5M]
b Describe how use cases are used to identify system requirements and explain with example? L2 CO2 [5M]

(OR)

- 4 a Discuss the importance of requirements gathering in the software development life cycle L2 CO2 [5M]
b Explain the concept of object-based requirements analysis and its role in modern software development L2 CO2 [5M]

UNIT-III

- 5 a Explain about use case diagram with suitable example? L3 CO3 [5M]
b What is generalization in UML and how does it help to simplify a class diagram and give an example? L3 CO3 [5M]

(OR)

- 6 a What are the several relationships in a class diagram, and how they are represented with multiplicities? L3 CO3 [5M]
b What are key components of a UML class diagram and build a use case diagram for online shopping system? L3 CO3 [5M]

UNIT-IV

- 7 a Describe the user interface design and explain how to evaluate the User Interfaces? L3 CO4 [5M]
b What are the difficulties and risks associated with modelling interactions and behaviours? L3 CO4 [5M]

(OR)

- 8 a What are the key usability principles that should guide the design of user interfaces? L3 CO4 [5M]
b Draw relevant state chart and activity diagram for Bank ATM transactions. L3 CO4 [5M]

(P.T.O)

UNIT-V

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|-------------|---|---|----|-----|------|
| 9 | a | What is the client-server architecture, and how does it enable distributed computing? | L3 | CO5 | [5M] |
| | b | Implement a model driven development for online banking system? | L3 | CO5 | [5M] |
| (OR) | | | | | |
| 10 | a | What techniques can be used to make effective design decisions in software architecture? | L3 | CO5 | [5M] |
| | b | Evaluate the Object Client-Server Framework, how to integrate it into existing client-server application? | L3 | CO5 | [5M] |
